How to design ERNIE rings

Step 1 – Install CAD Software

Install a CAD software package capable of creating and editing parametric 3D models. The ERNIE rings were designed using **Autodesk Inventor Professional**.

Students can obtain a free educational license from Autodesk:

https://www.autodesk.com/education/edu-software/overview#INVPROSA

Although Autodesk Inventor is used throughout this guide, the same design workflow can be implemented using SolidWorks, AutoCAD, or other equivalent CAD software.

Step 2 – Review the Ring Design Parameters

Before creating a ring model, review the optimization results to determine the dimensions and magnet configuration of the ring to be designed.

For example, Ring #288 has:

Inner bore diameter: 80 mm

Outer ring diameter: 160 mm

Outer shim tray diameter: 190 mm

Configuration_Number	Band_Number	Band_Radii_Gap_mm	Magnet_Space_mm	Band_Separation_mm	Band_1_Radius_mm	Band_1_Magnet_Count
Ring_321	1	0	8.82	16.95	65.43	16
Ring_288	1	0	11.76	15.29	63.77	14
Ring_251	1	0	8.82	13.63	62.11	15
Ring_42	1	0	10.2941	3.6610	52.15	12

Total number of magnet used = (Ring_321(16 magnets), Ring_288(14 magnets) *2,* Ring_251(15 magnets)*2 and Ring_42(12 magnets)*2). = 16+28+30+24 = 98 magnets.

Step 3 – Create a New CAD Part

Launch Autodesk Inventor and create a new metric part by selecting:

File → New → Templates → Metric → Standard (mm).ipt → Create

Using the metric template ensures that all dimensions are defined in **millimetres**.

For improved visibility, the workspace background may be changed by navigating to:

File → Options → Colors → Sky

Select **Apply** to save the changes.

Step 4 – Prepare the Sketching Workspace

When a new part is created, the reference planes are hidden by default. To make them visible, expand **Origin** in the Model Browser, right-click **Planes**, and select **Visible**. Repeat this for the **YZ**, **XZ**, and **XY** planes.

Next, right-click the **XY** plane and select **Look At** (or navigate to the **Sketch** tab and select the **Perpendicular** view) to orient the workspace normal to the sketch plane before beginning the design.

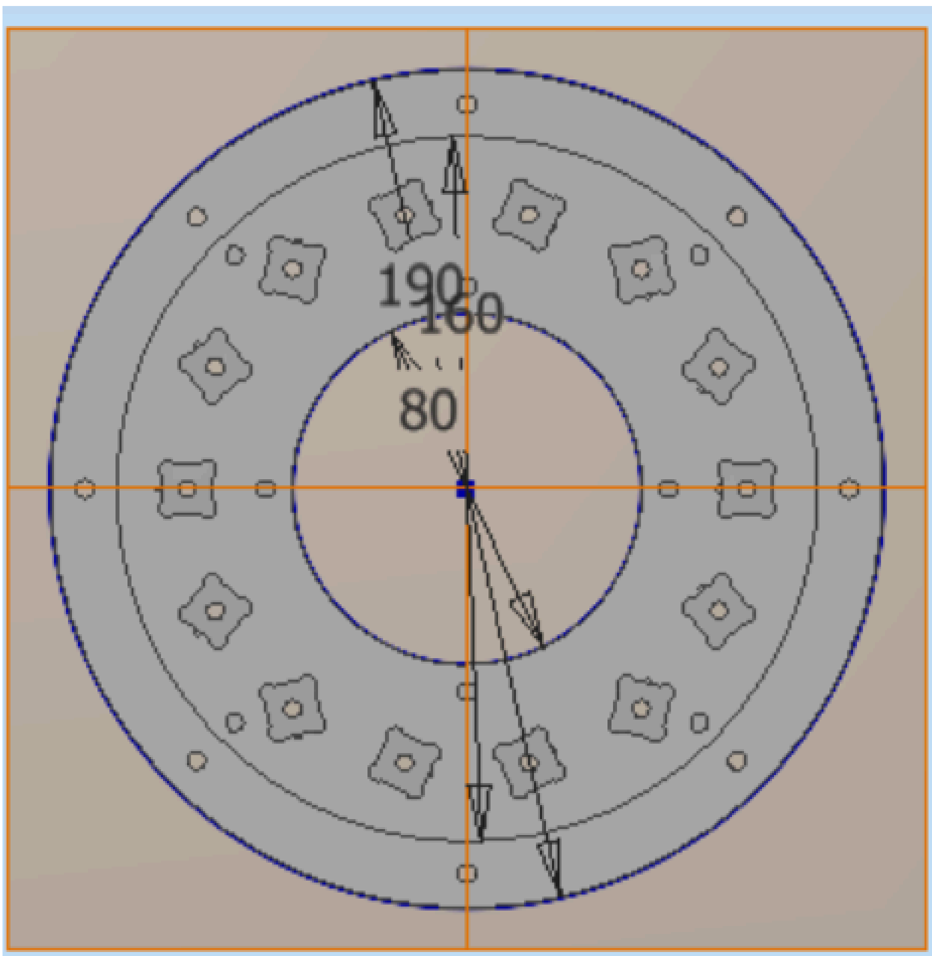
Step 5 – Create the Ring Geometry

Create a new sketch on the **XY** plane. From the **Sketch** tab, select **Center Point Circle** and draw three concentric circles, all originating from the center point.

- Set the **inner bore** diameter to **80 mm**.

- Set the **outer ring** diameter to **160 mm**.
- Draw a third concentric circle with a diameter of **190 mm** to define the outer boundary for the shim trays.

These three circles establish the primary geometry of the ERNIE ring.



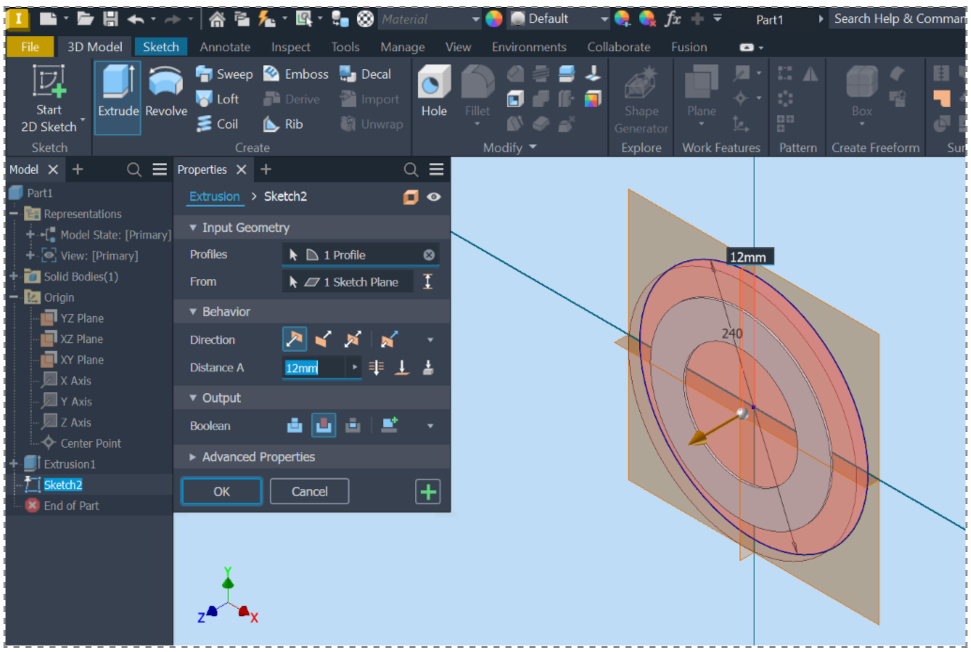
Step 6 – Extrude the Ring Geometry

After completing the sketch, select **Finish Sketch** to create the initial ring features.

First, select the inner circular region and apply an **extrusion of 14 mm** to create the inner section of the ring. After completing this extrusion, right-click the original sketch in the Model Browser and select **Visible** to display it again.

Next, select the outer circular region and apply an **extrusion of 12 mm** to create the remaining section of the ring.

The resulting geometry forms the stepped profile of the ERNIE ring, with a **14 mm inner section** and a **12 mm outer section**.



Step 7 – Create the Magnet Band Reference Circle

The magnet band dimensions are obtained from the optimized field results for each ring. For example, **Ring #288** has a specified magnet band radius of **63.77mm**.

Create a reference circle representing the magnet placement band by setting its diameter to:

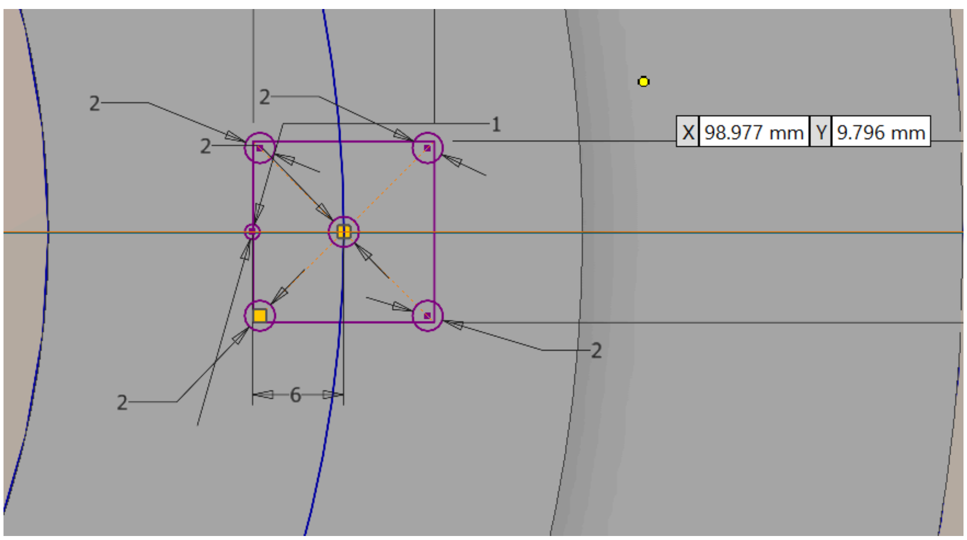
2 × 63.773 mm = 127.546 mm

This reference circle defines the location where the magnet slots will be positioned within the ring.

Step 8 – Create the Magnet Block

Select **Sketch 5**, which contains the magnet band reference circle, and choose **Edit Sketch** to begin creating the magnet block.

Select the two side reference rectangles and draw a horizontal construction line passing through the center of the ring and intersecting the magnet band circle at **180°**. This line establishes the reference position and orientation for placing the first magnet block.



The dimensions of the neodymium magnets are measured as 11.8 mm × 11.8 mm × 11.8 mm. Therefore, dimension the magnet block as an 11.8 mm × 11.8 mm square to match the magnet dimensions.

Add a small reference point on the side of the square facing the center of the ring. This point defines the magnet rotation direction during placement within the Halbach array. Position the center of this reference point 6 mm from the center of the square and set its diameter to 1 mm.

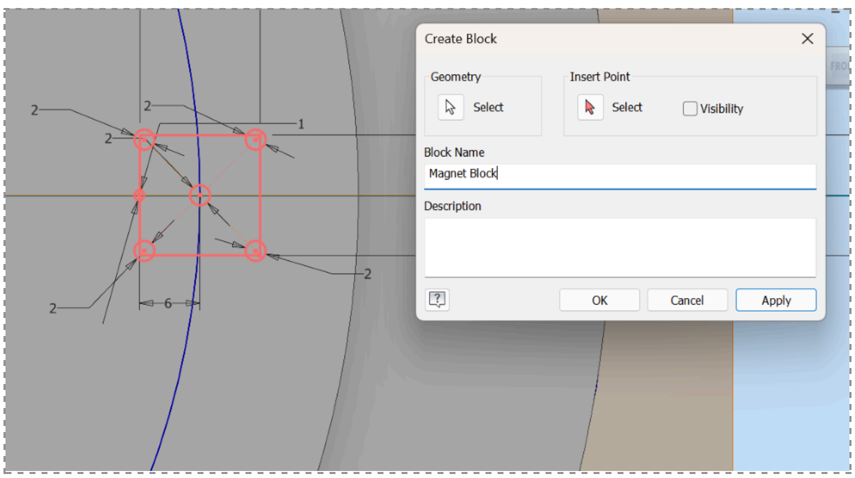
Add four additional circles at the corners of the square. Set each corner circle diameter to 2 mm. These corner features provide clearance and allow the magnet to fit easily within the slot during assembly.

Add a circular opening at the center of the square and set its **radius to 2 mm**. This feature provides access for easier removal of the magnet from the slot if replacement or adjustment is required.

To convert the completed magnet slot geometry into a reusable block, navigate to:

Model Browser → Block → Create Block

Select all lines, circles, and reference points associated with the newly created magnet block.

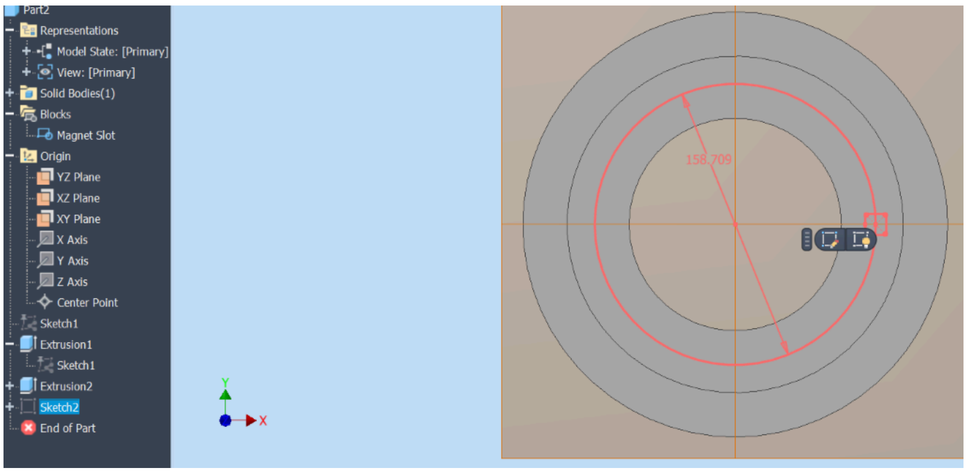


Select the **Insert Point** option and define the **center of the magnet block** as the insertion point. This establishes the reference location used when placing the magnet block around the magnet band.

Assign the block name **“Magnet Block”** and select **Apply** to create the block. After creation, the center points of the circles and rectangles within the block should change to **white**, indicating that the block has been successfully defined.

Select the green check mark to complete the block creation process.

To improve visibility, hide any unnecessary sketches and reference features, leaving only the magnet block visible. Finally, finish the sketch and hide it before proceeding to the next design step.



While remaining in **Sketch Mode**, create a new magnet band reference circle and set its diameter according to the design parameters. For **Ring #288**, the magnet band diameter is **127.546 mm**.

Place a **Magnet Block** onto the reference circle. Initially, moving the block will cause the entire block to move freely. To constrain its rotation, draw a construction line from the **center of the ring** to the corresponding point on the magnet band circle. Position the magnet block at the intersection point between this line and the band circle. The center point of the magnet block should turn **green**, indicating that it is correctly constrained.

Verify the constraint by moving the magnet block. The block should rotate smoothly around the ring center through **360°** while maintaining its position on the magnet band.

From the optimization results file, identify the required number of magnets for the selected ring. For example, **Ring #288 requires 14 magnets**.

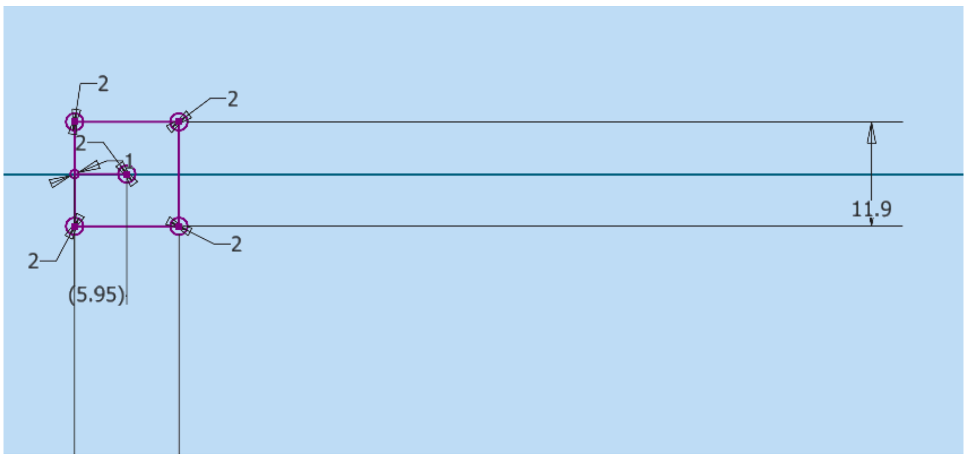
To distribute the magnet blocks around the ring, use the **Circular Pattern** tool:

1. Select the horizontal construction line connecting the ring center to the magnet block center as the reference direction.
2. Select the center of the ring as the rotation axis.

3. Define the required number of pattern instances based on the magnet count.

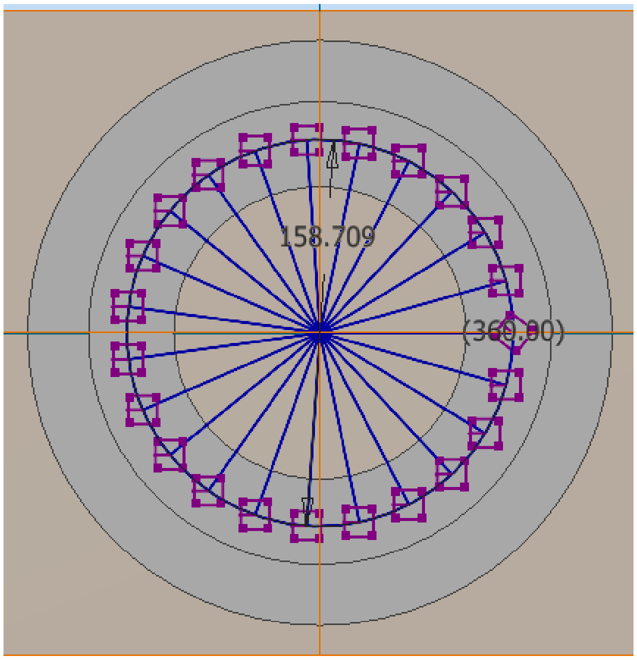
For Ring #288, create a pattern of **7 magnet blocks**. The remaining magnets are then placed by mirroring the arrangement to complete the ring configuration.

When placing each magnet block, ensure that the **insert point (center point)** turns green, confirming that the block is correctly positioned and constrained.



While still in **Sketch Mode**, select **Edit Block** to modify the magnet block geometry.

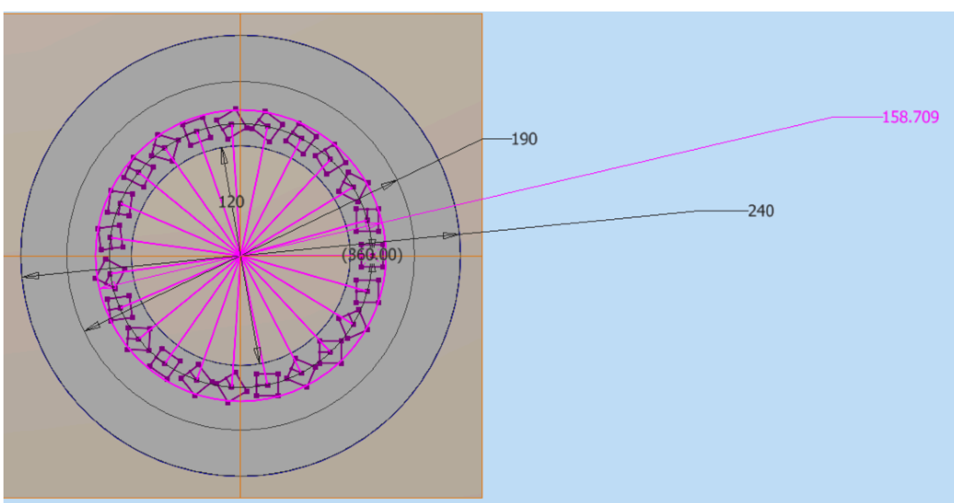
Draw a reference line connecting the **center of the magnet block** to the **center of the small 1 mm diameter reference circle** positioned 6 mm from the center of the magnet slot. This line defines the magnet orientation and serves as a reference for the rotation direction of the magnets within the Halbach array.

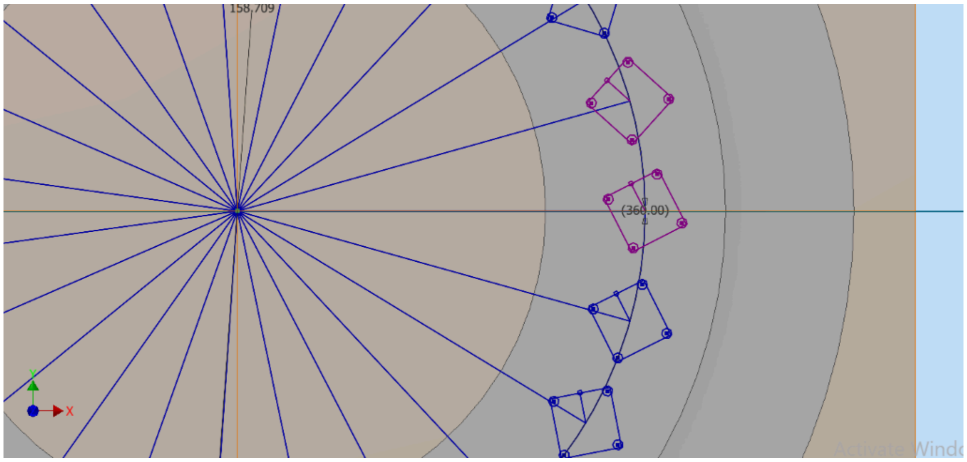


Step 9 – Define the Angular Orientation of Magnet Slots

Before applying angular dimensions to the magnet slots, slightly tilt the slots to prevent them from automatically constraining or snapping to existing geometry. This ensures that the applied dimensions are interpreted as **angular constraints** rather than linear measurements.

This step prepares the magnet slots for accurate definition of their rotational positions within the Halbach array.





Step 10: Set Dimensions for the magnet slots.

To simplify the dimensioning process, ensure that all magnet slots are **purple**, indicating that they are flexible. This allows their dimensions and angular orientations to be modified. If the magnet slots appear **blue**, they are already fixed and cannot be adjusted.

The angular separation between the magnet slots is determined using the Halbach array formula:

360(2)/N

where **N** represents the number of magnets in the ring.

For example, for the **120 mm inner bore diameter rat scanner**, the first ring designed was the middle ring (**Ring #842**). Based on the results obtained from the optimization algorithm, this ring contains **20 magnets**.

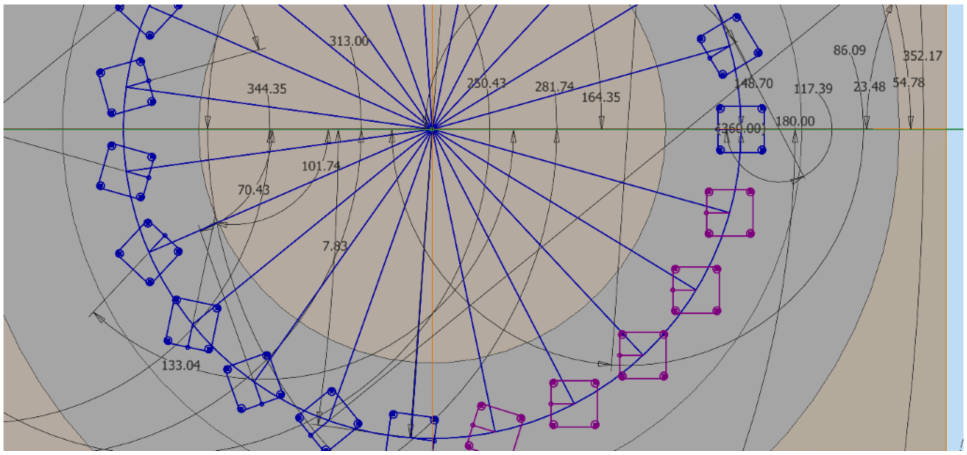
The angular spacing is calculated as:

360(2)/20 = 36°

This means that there is an angular difference of **36°** between each magnet slot.

Since the magnet slots are rotated in the **counterclockwise direction**, this value is entered as a negative angular dimension in the CAD software.

After applying the required angular dimension to a slot, its orientation becomes fixed. Fixed magnet slots appear **blue**, while flexible slots remain **purple**.

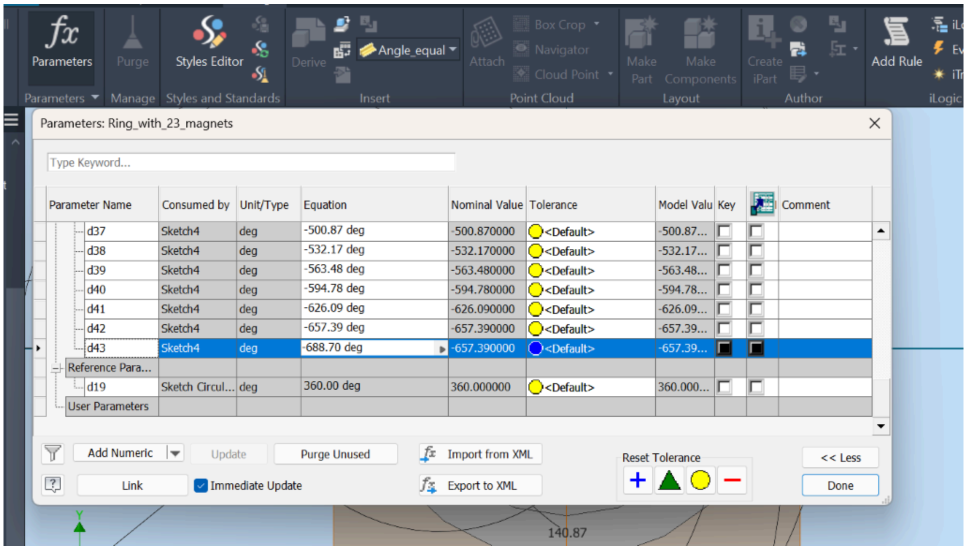


After applying all angular dimensions, all magnet slots should appear **blue**, indicating that they are fully constrained.

Next, navigate to:

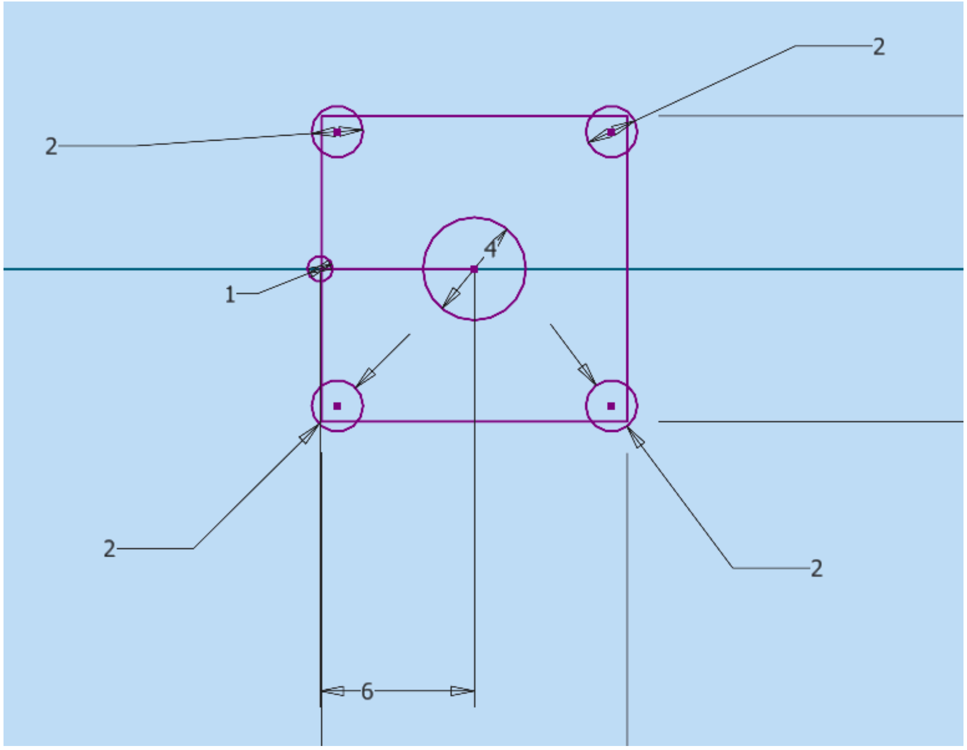
Manage → Parameters

Review the listed angular parameters to verify that all magnet slot orientations have been correctly defined. If any incorrect values or errors are identified, make the necessary corrections before saving the final design.



Step 11: Modify the Magnet Block for Magnet Removal

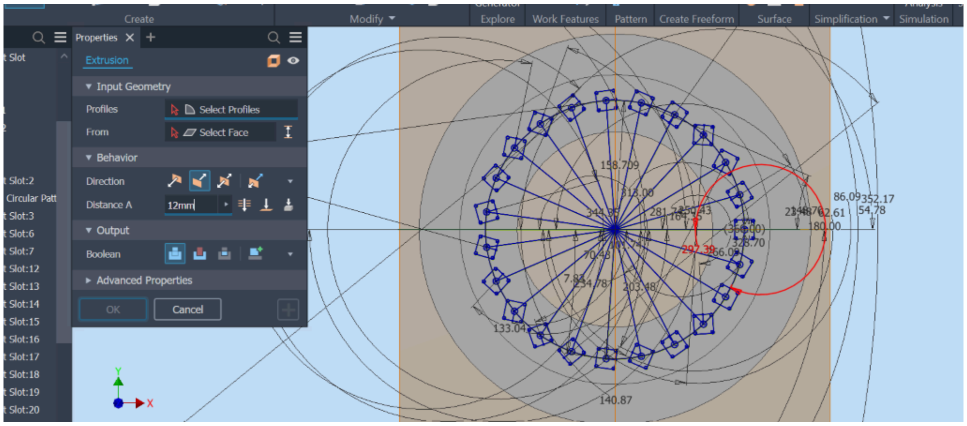
Edit the **Magnet Block** and add a circular opening at the center of the block. This opening provides access for pushing the magnet out of the slot during removal or replacement, making the assembly and maintenance process easier.



Step 12: Extrude the Magnet block

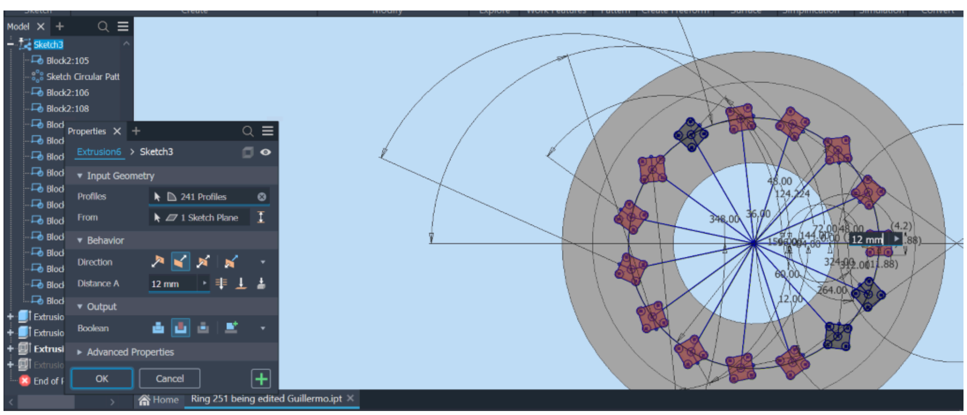
The measured dimensions of the neodymium magnets are **11.88 mm × 11.88 mm × 11.88 mm**. For the CAD design, these dimensions are rounded to **12 mm** to define the magnet slot depth and ensure sufficient clearance for magnet insertion.

Select the magnet slot regions and apply an extrusion of **12 mm** to create the cavities where the magnets will be placed within the ring.

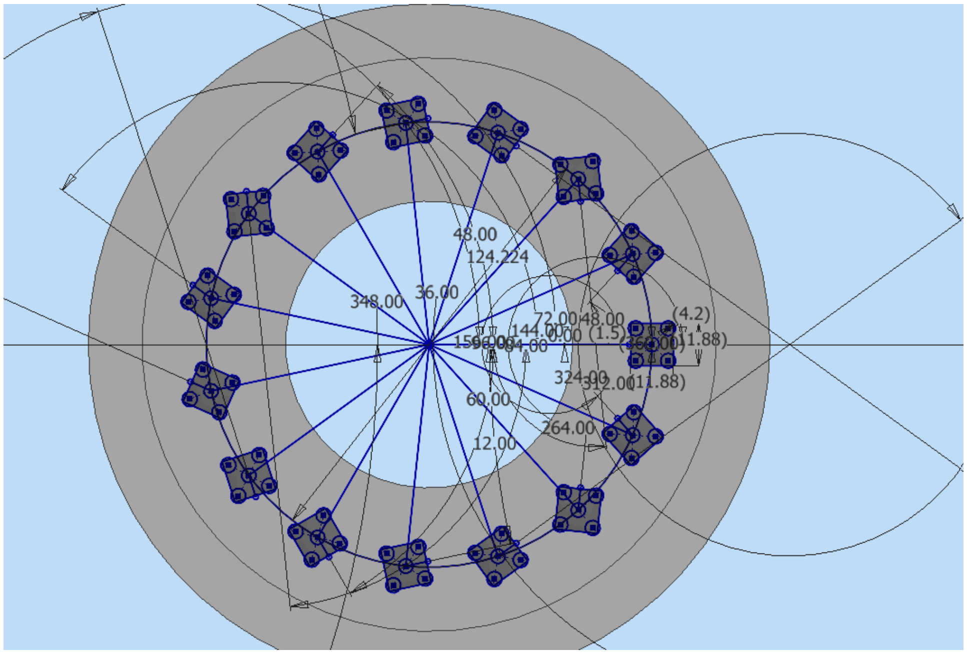


For this extrusion, select each magnet slot individually along with the four pressure relief circles located at the corners of each slot.

Once all required profiles are selected, proceed with the extrusion operation to create the magnet cavities and the associated clearance features.



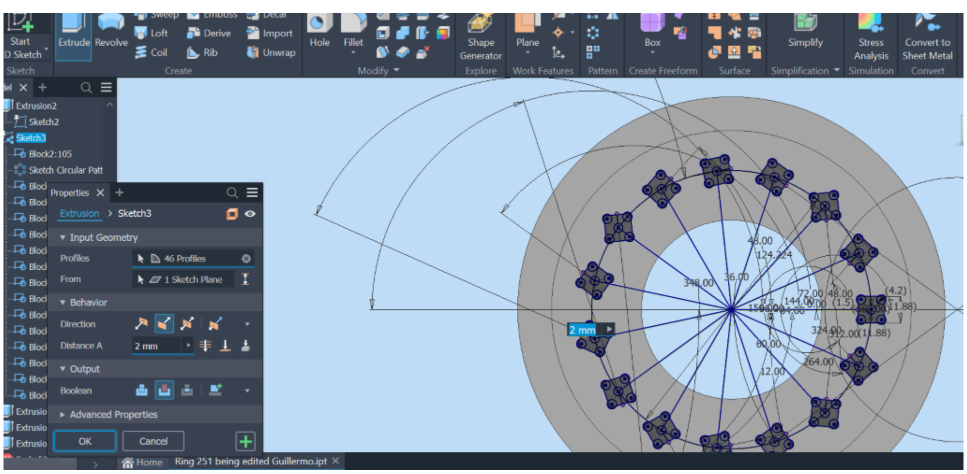
After extrusion (In Sketch Mode) the ring looks like this one below:



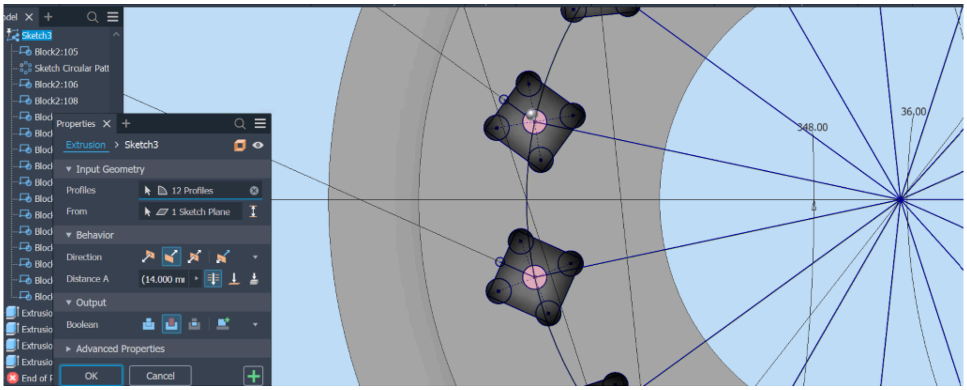
Step 13: Extrude the 1.5 mm Diameter Holes

Next, create the small **1.5 mm diameter holes** used as additional clearance features in the magnet slots.

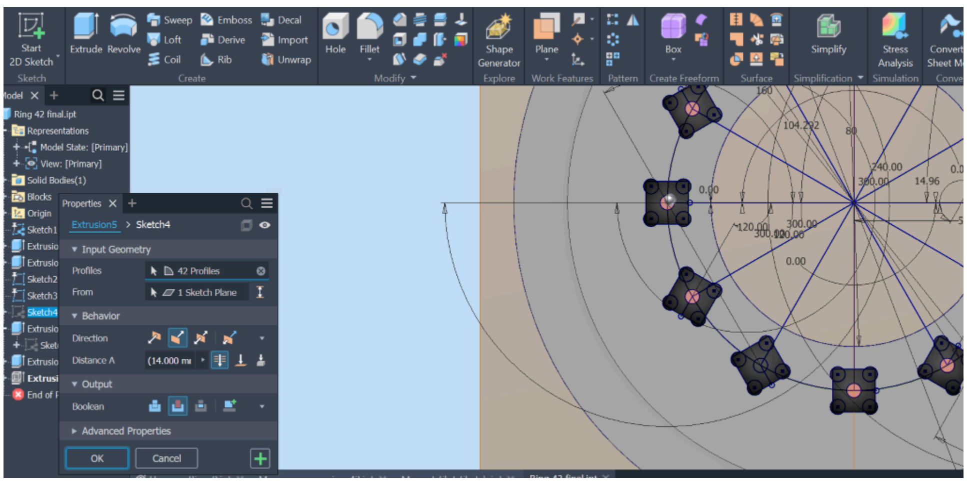
Select **Edit Sketch** to access the hole profiles, then select all the 1.5 mm diameter circles. Apply an extrusion depth of **2 mm** to create these features.

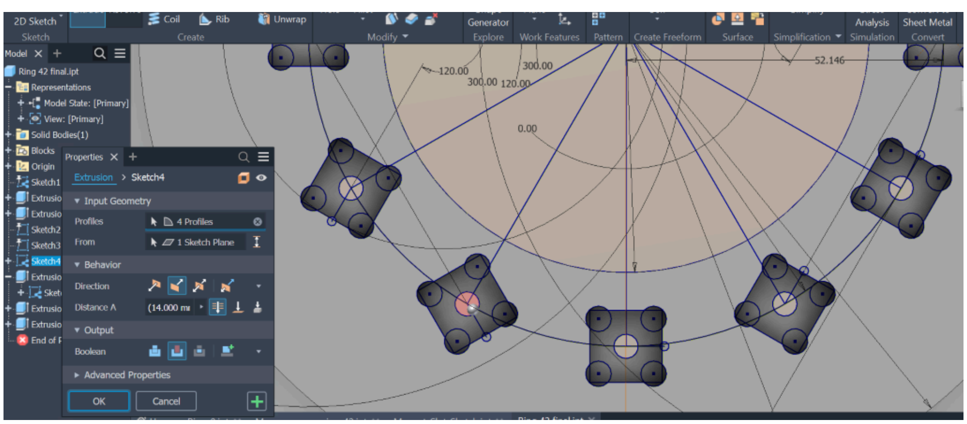


Step 13: Extrude cut through all the 4mm holes.

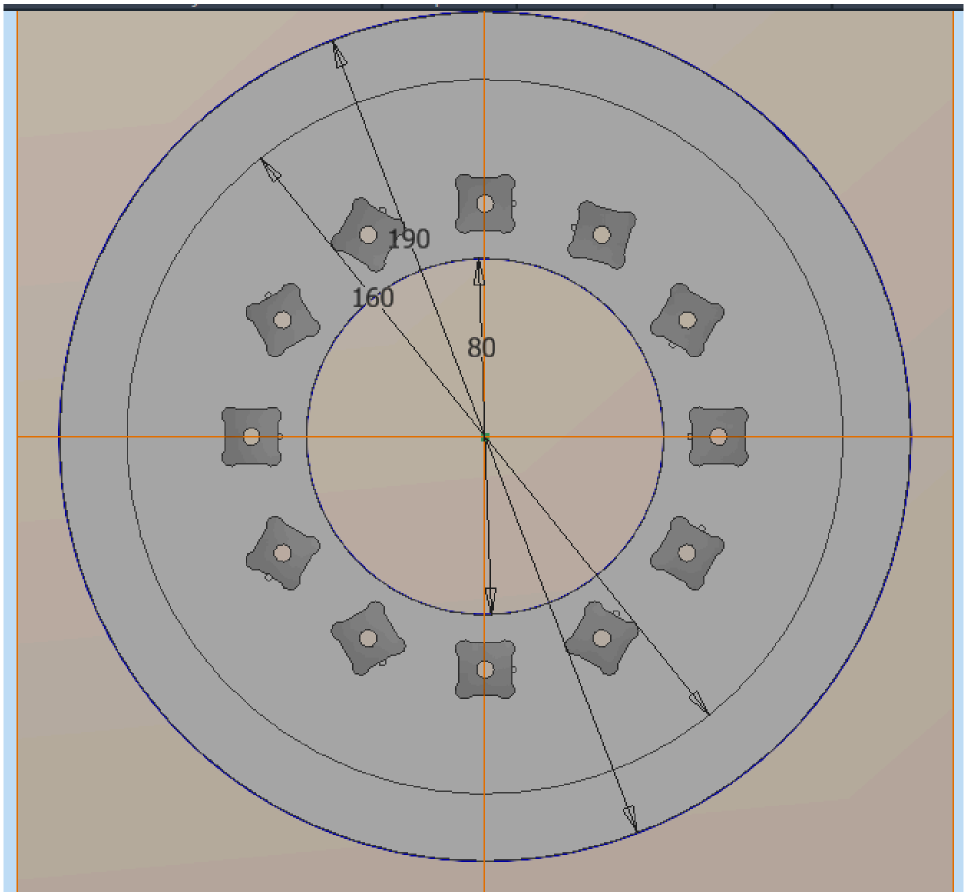


Ensure to cut through all .





After this step, the Rings appear like the one below:



Step 14: Create Magnet Removal Holes and Mounting Features

The magnet slots are created with an **extrusion depth of 12 mm** to accommodate the neodymium magnets. The central hole used for magnet removal is created using an **extrude cut through all**, allowing access to push the magnet out of the slot when required.

The hole dimensions for the mounting features are defined as follows:

- **M3 screw holes:** 4.2 mm diameter
- **M4 brass rod holes:** 4.3 mm diameter

To create these features:

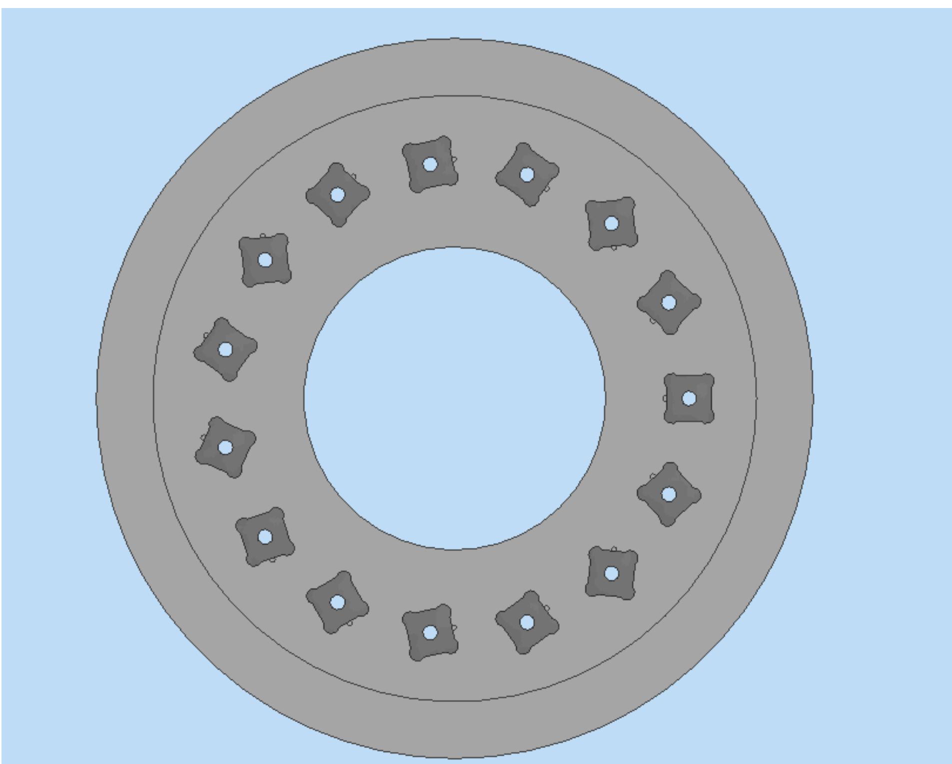
1. Make the previously drawn hole sketches visible.
2. Select the **Extrude** command and create the required hole features.

3. Create additional small circular sketches for the M3 screws used to secure the lids onto the rings.
4. Create the M4 circular holes through which the brass rods will pass.
5. Use **Extrude Cut → Through All** to create the holes through the entire ring thickness.

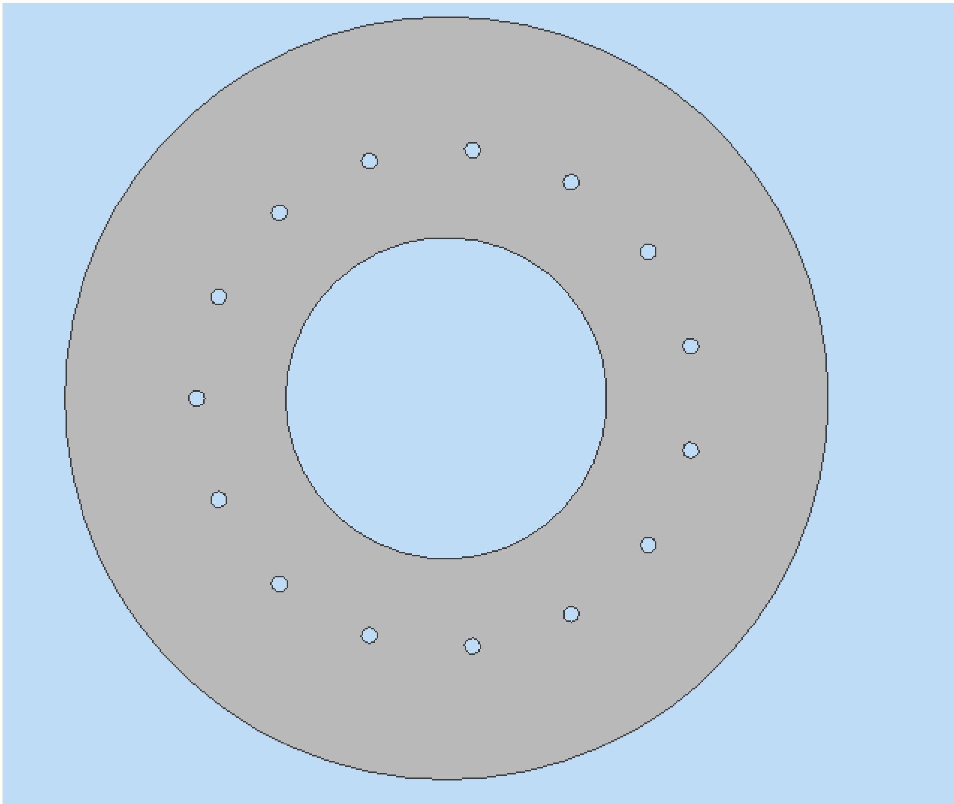
After completing these steps, the ring geometry should contain the magnet slots, magnet removal holes, lid mounting holes, and brass rod openings.

The figure below shows the final appearance of **Ring #251 for the 50 mT small scanner** before adding the remaining holes for the brass rods and lid fastening screws.

Front view



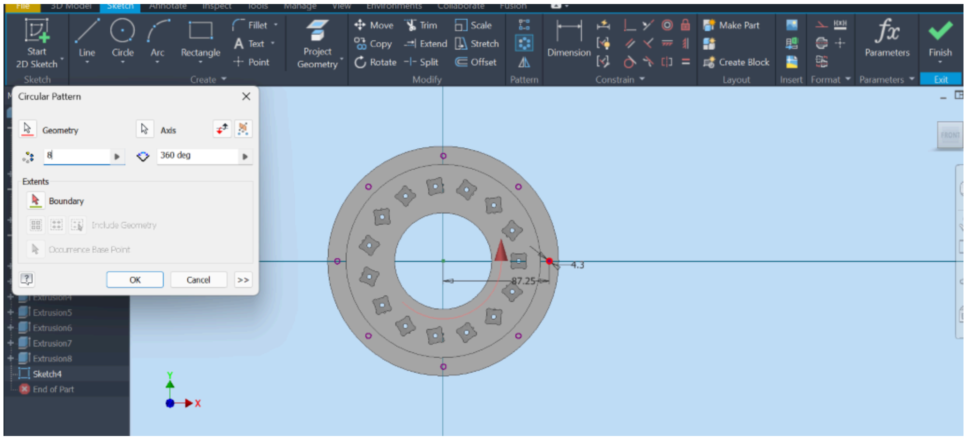
Back view



Step 15 – Create Brass Rod Holes

To create the holes for the brass rods:

1. Create a new sketch on the ring surface.
2. Draw a circle for the brass rod opening and set its diameter to **4.3 mm**.
3. Dimension the circle immediately after creating it to ensure the correct size is maintained.
4. Use the **Circular Pattern** tool to replicate the hole around the ring, creating a total of **8 equally spaced holes**.
5. Select **Extrude Cut → Through All** to create the brass rod holes through the entire ring.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Extrusion5
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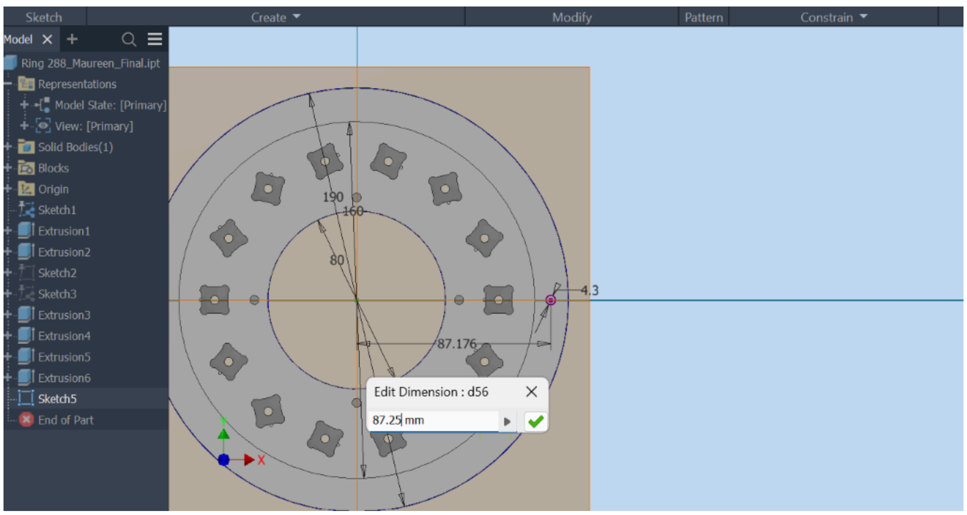
Extrusion7
Extrusion8
Sketch4
End of Part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Pattern
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Model
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Representations
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View: [Primary]
Solid Bodies(1)
Blocks
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Extrusion4
Extrusion5
Extrusion6
Sketch5
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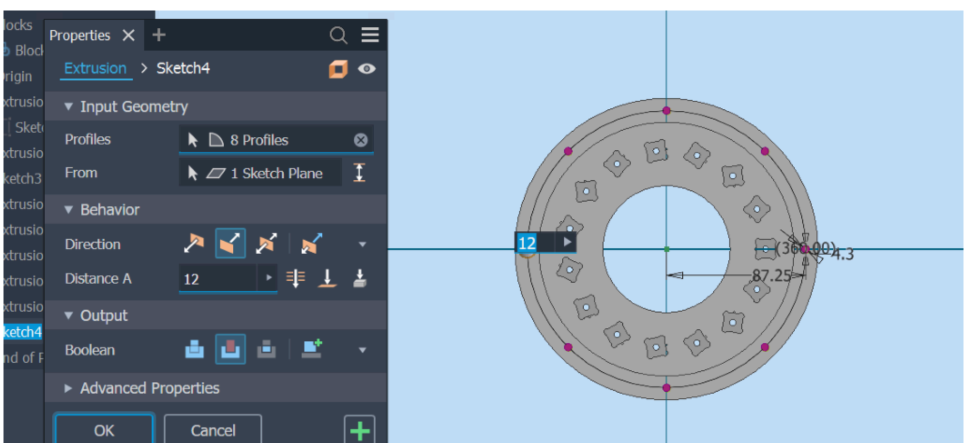
Properties
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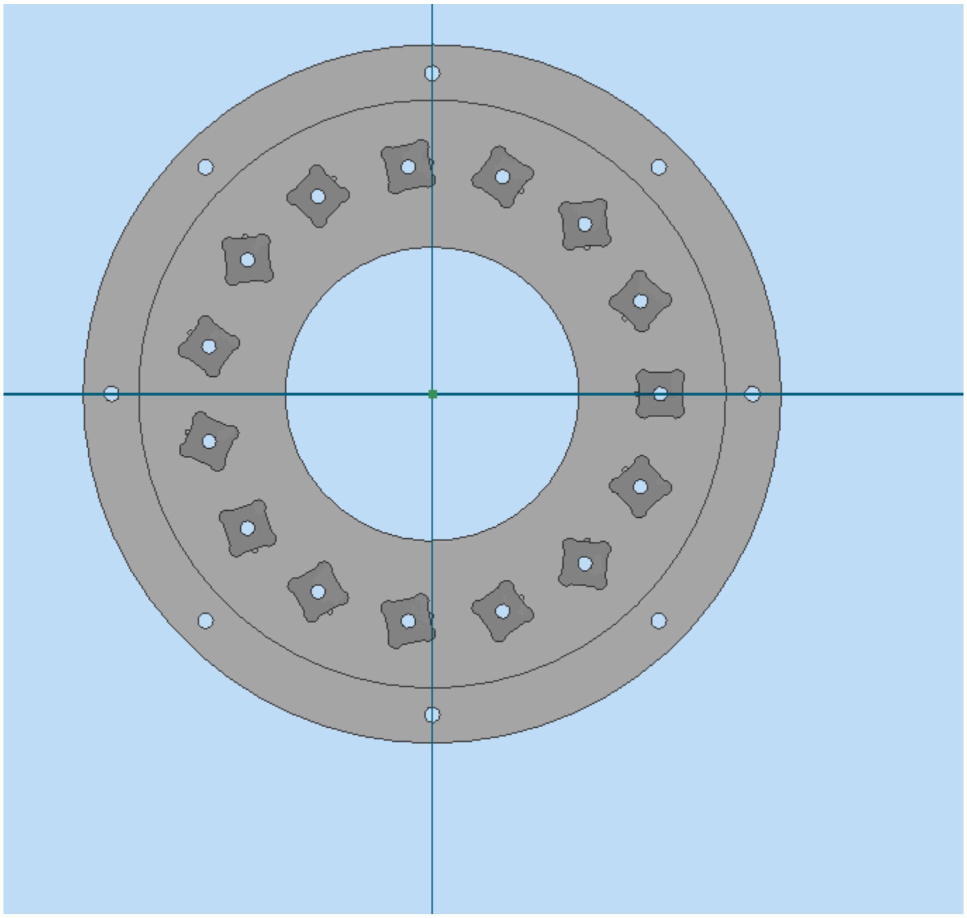
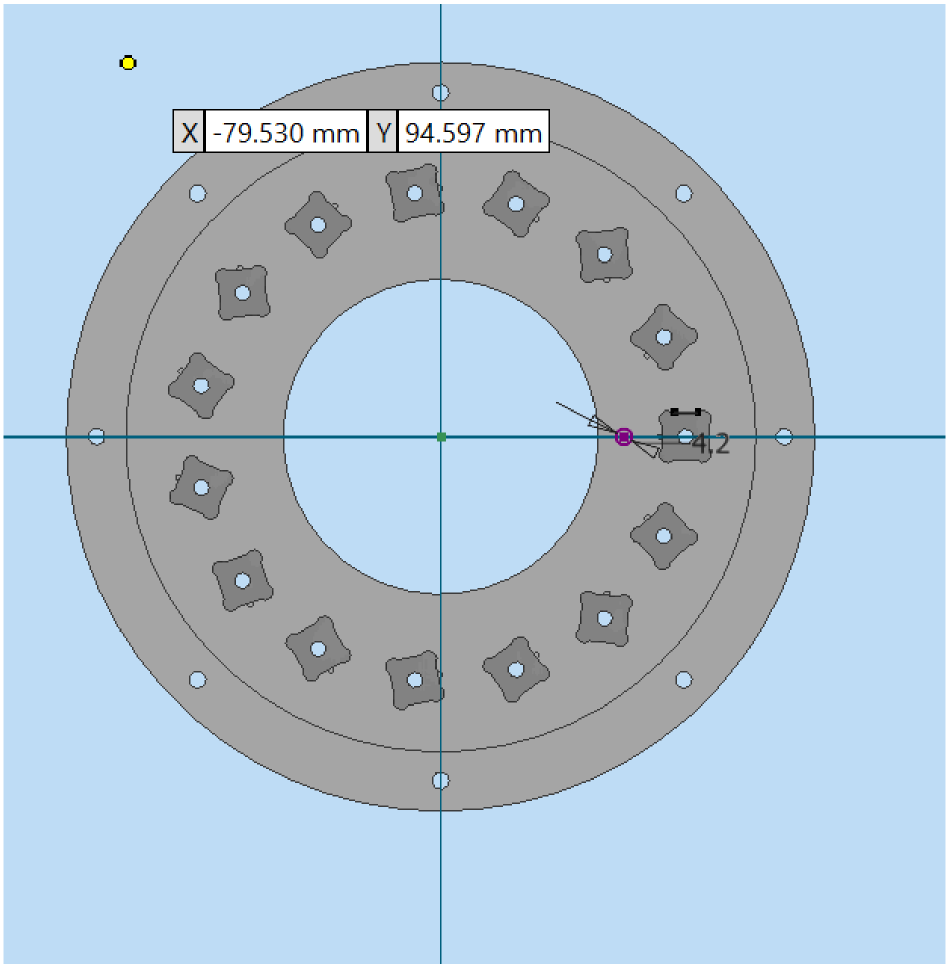
Final result



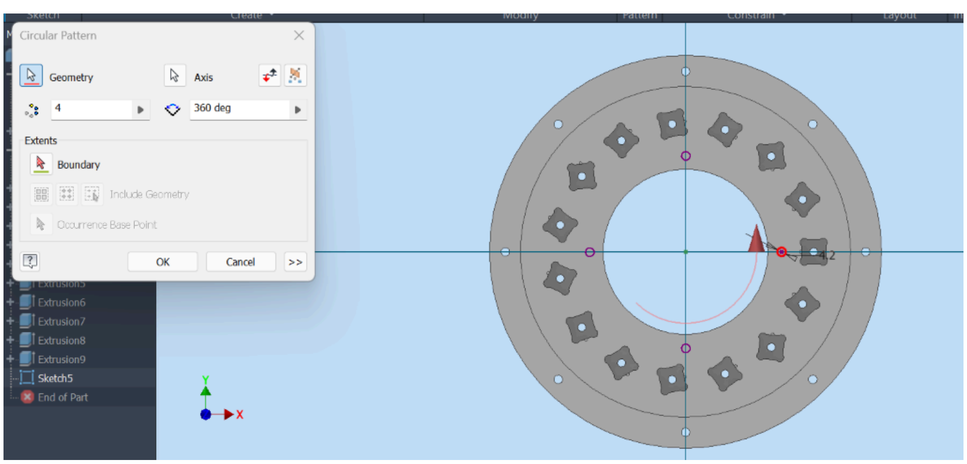
tep 16 – Create Heat-Set Insert and Lid Screw Holes

After creating the brass rod holes, add the holes required for installing the heat-set inserts and securing the lids.

1. Create a new sketch on the interior surface of the ring.
2. Draw the holes for the M3 screws and set their diameter to **4.2 mm**.
3. Position the holes according to the ring design specifications.
4. Ensure that the holes do not overlap with the magnet slots, as this may affect the structural integrity of the ring.
5. Apply **Extrude Cut → Through All** to create the holes.



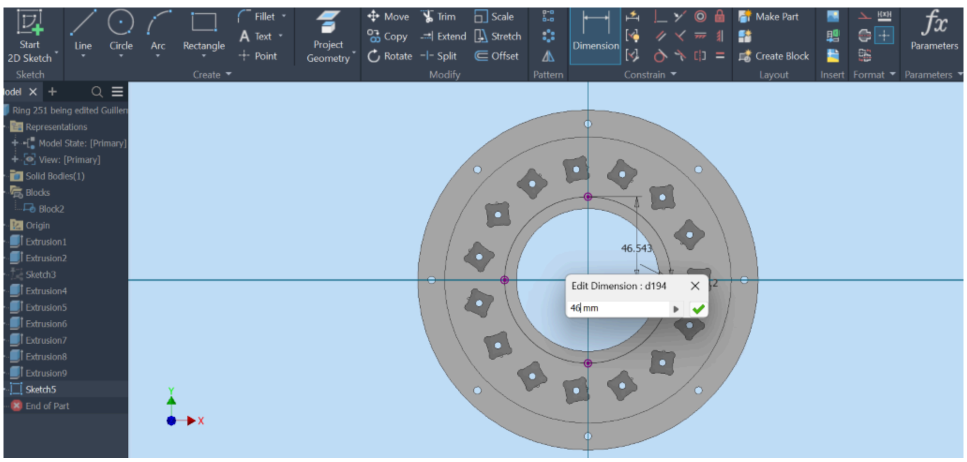
Final Result

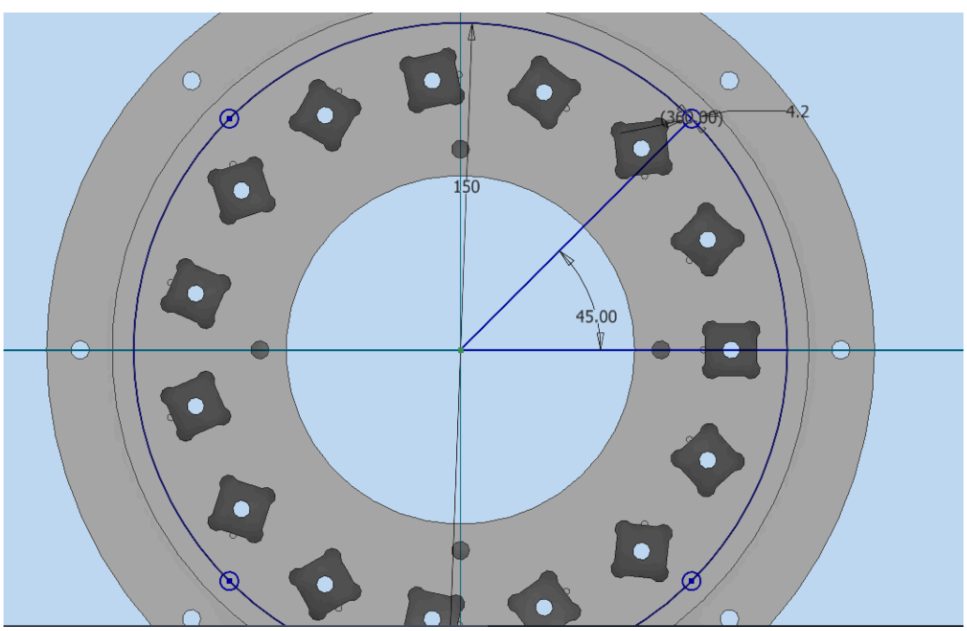


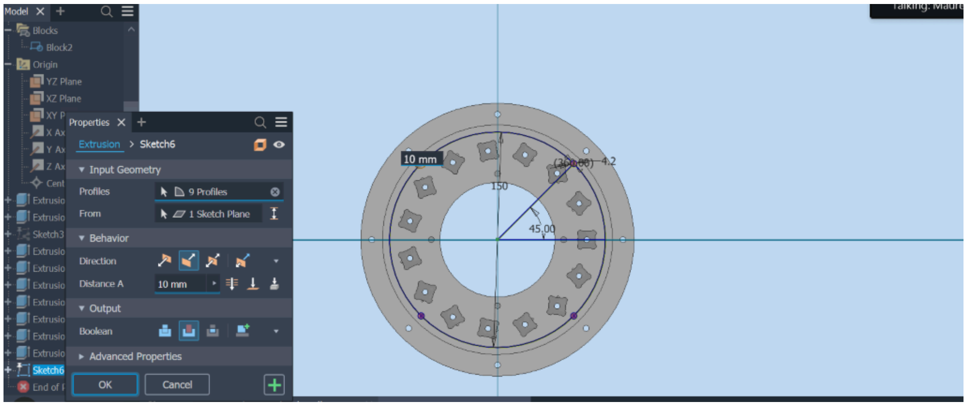
Step 17 – Create the Second Set of Inner Lid Screw Holes

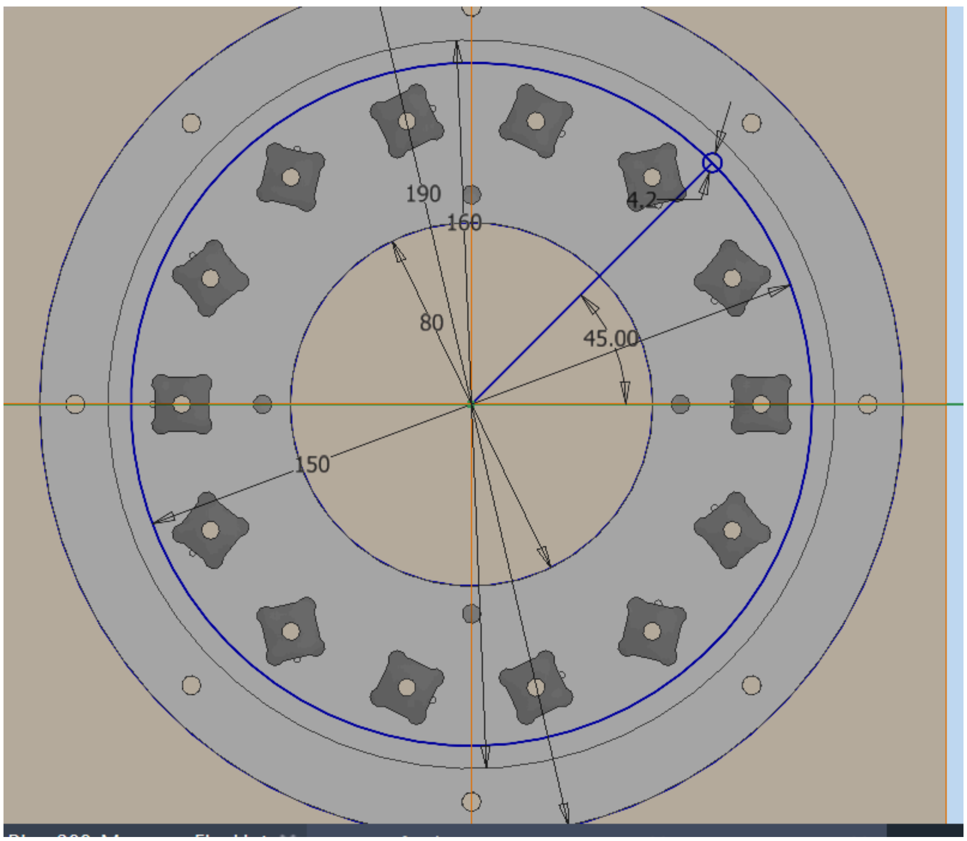
For the second set of inner screw holes:

1. Draw a reference circle with a diameter of **150 mm**.
2. Add M3 screw holes with a diameter of **4.2 mm**.
3. Position the holes at **45° intervals** around the ring.
4. Verify that the hole locations do not interfere with the magnet slots before creating the cuts.

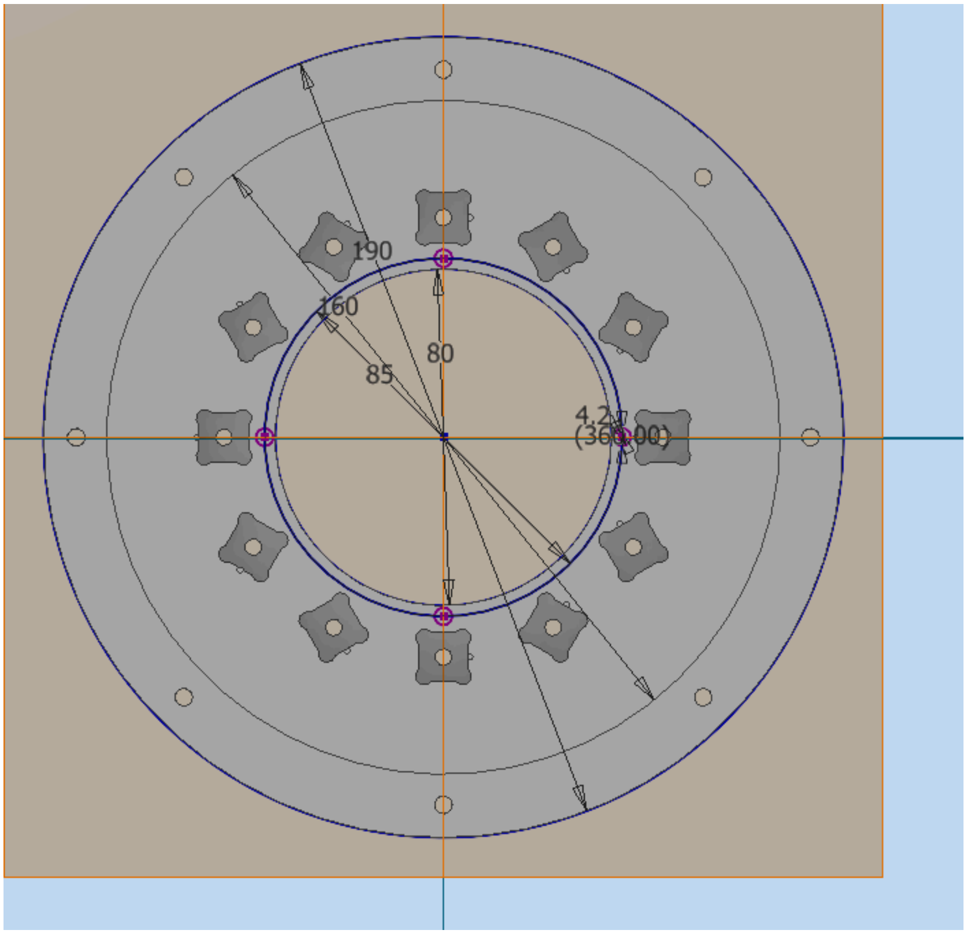








NOTE: While creating holes for insertion of the heat set inserts and then screws to tighten lids onto the rings, keen attention should be taken so that the holses cannor overlap with the magnet slots.

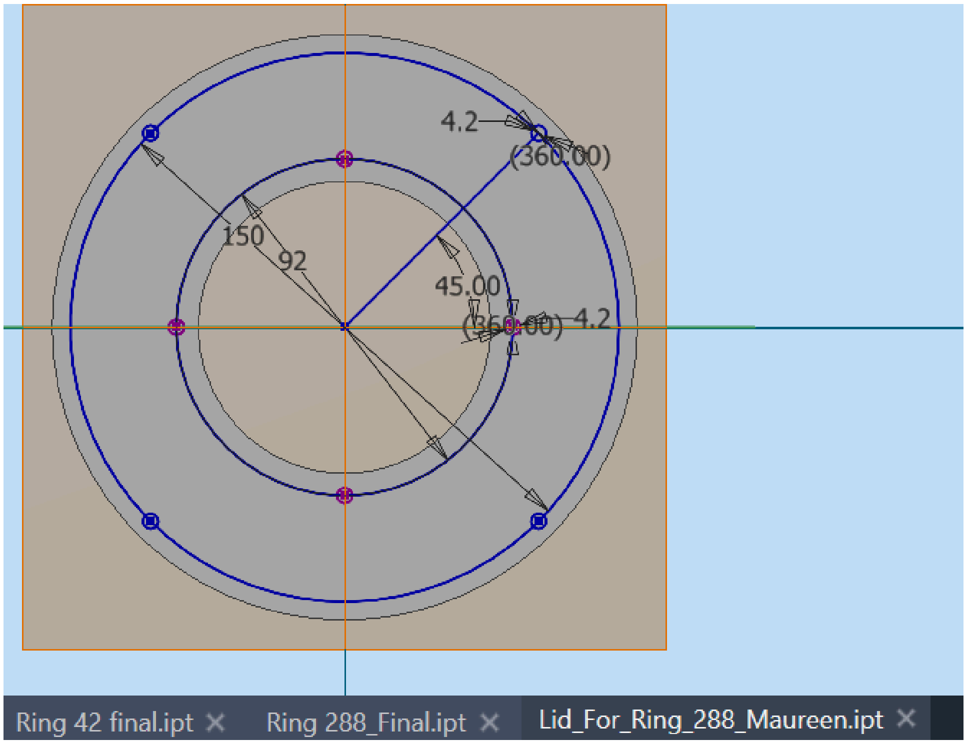


Step 18 – Ring #321 Design Considerations

Ring #321 requires a modified screw-hole arrangement because it is positioned further toward the exterior of the Halbach array.

For this ring:

- The reference circle immediately outside the magnet band is set to a diameter of **154 mm** to provide sufficient space for lid fastening holes.
- The corresponding lid is designed using the same dimensions to ensure proper alignment and fit.



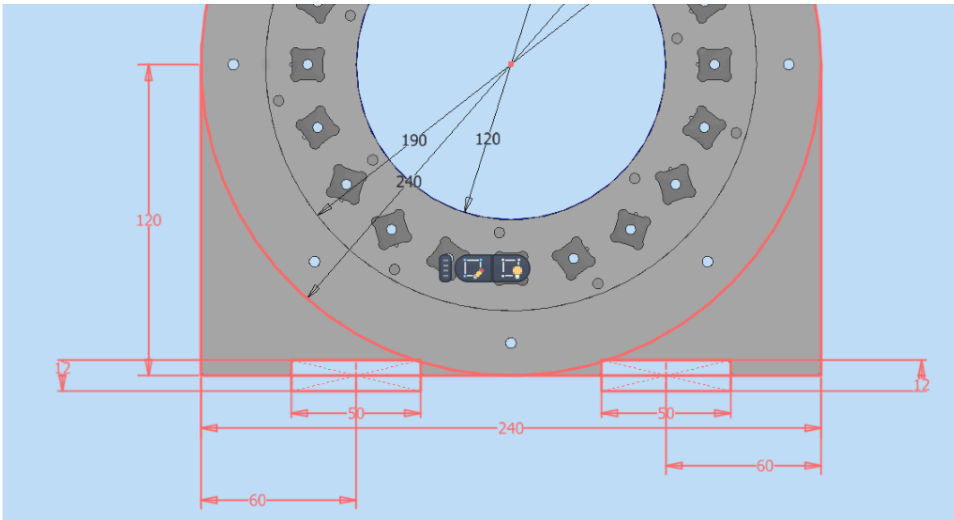
Step 19 – Design the Ring Supports for Ring #42 and Ring #321

The support structures for **Ring #42** and **Ring #321** are designed using the same approach. Since these rings are scaled versions of the **120 mm inner bore diameter rat brain scanner**, the support dimensions are adjusted according to the corresponding ring geometry.

1. Create a **new sketch** on the ring surface.
2. Define the support geometry using the required dimensions based on the scanner design parameters.
3. Apply an **extrusion of 12 mm** to create the support structure.
4. Apply **fillets** to the edges of the support to improve the mechanical finish and reduce sharp edges.
5. Adjust and test different design parameters to achieve the desired support geometry and ensure proper fit.

The final support design should be verified after extrusion and filleting to confirm that it matches the intended ring configuration.

New sketch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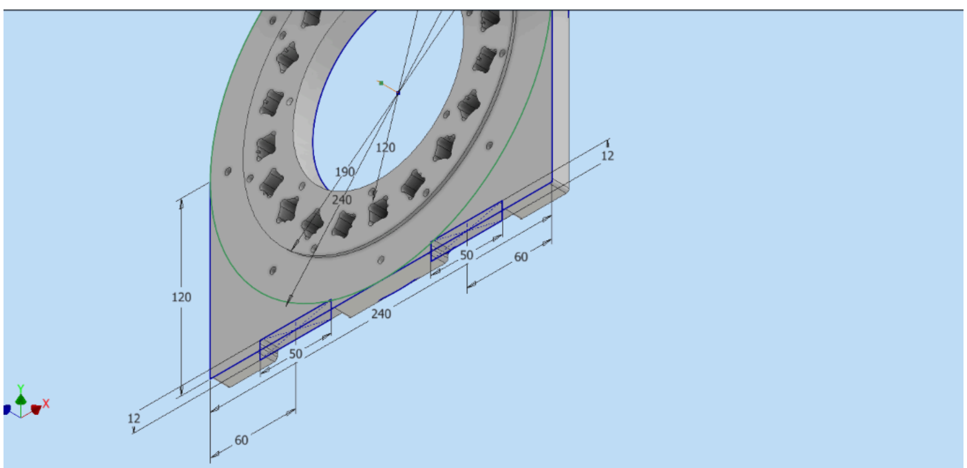
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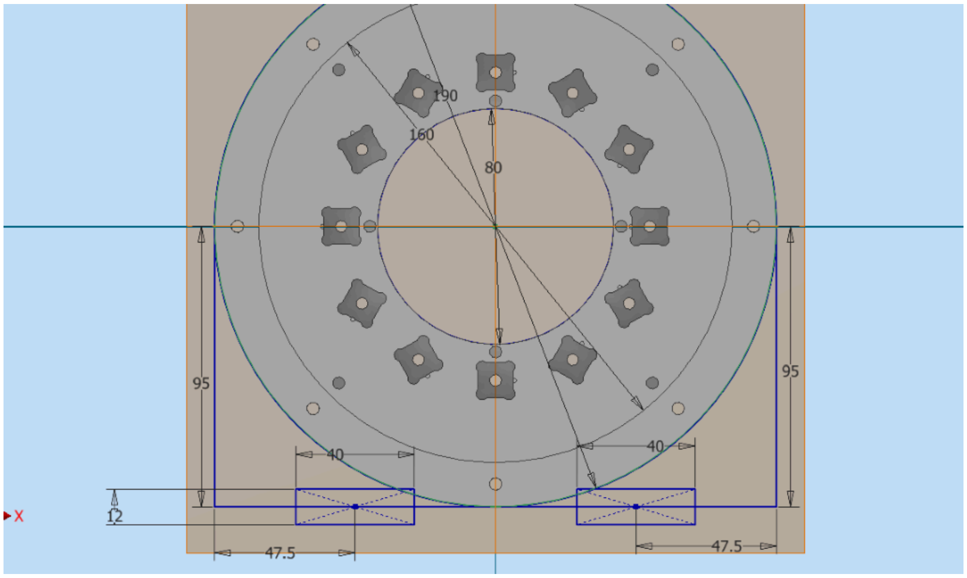
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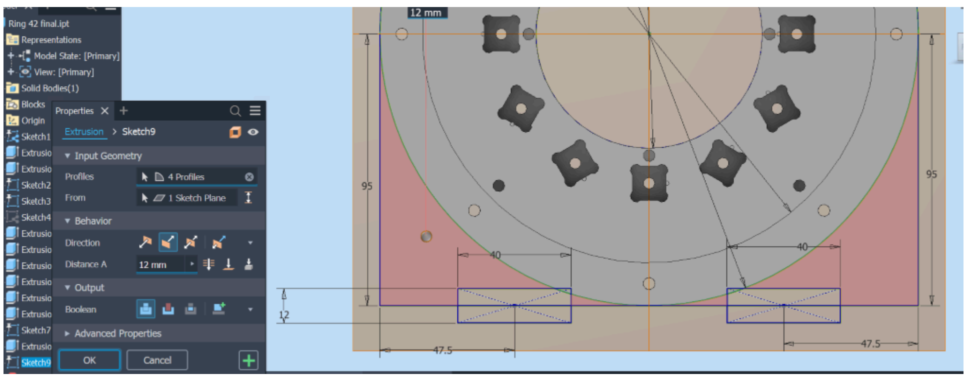
47.5


Ring 42 final.ipt
Representations
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View: [Primary]
Solid Bodies(1)
Blocks
Origin
Properties
Extrusion
Sketch9
Input Geometry
Profiles
4 Profiles
From
1 Sketch Plane
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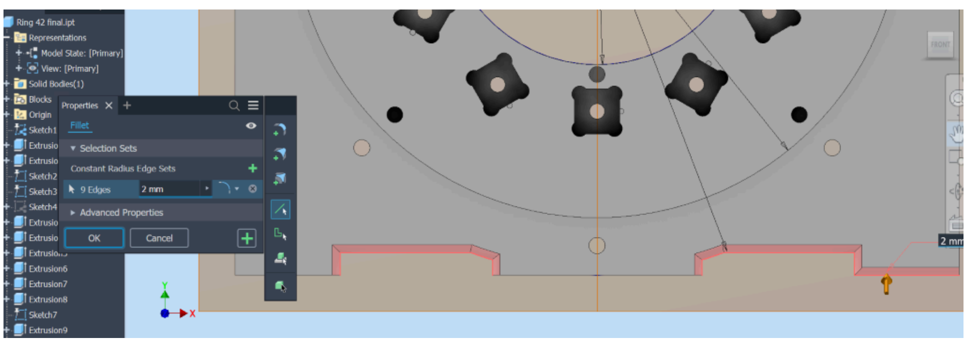
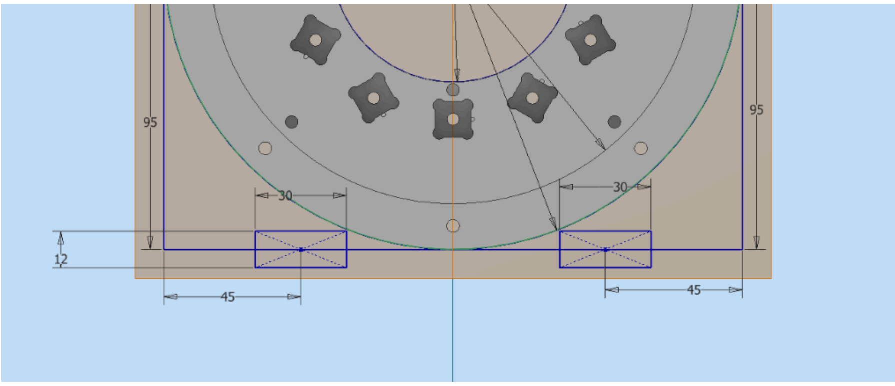
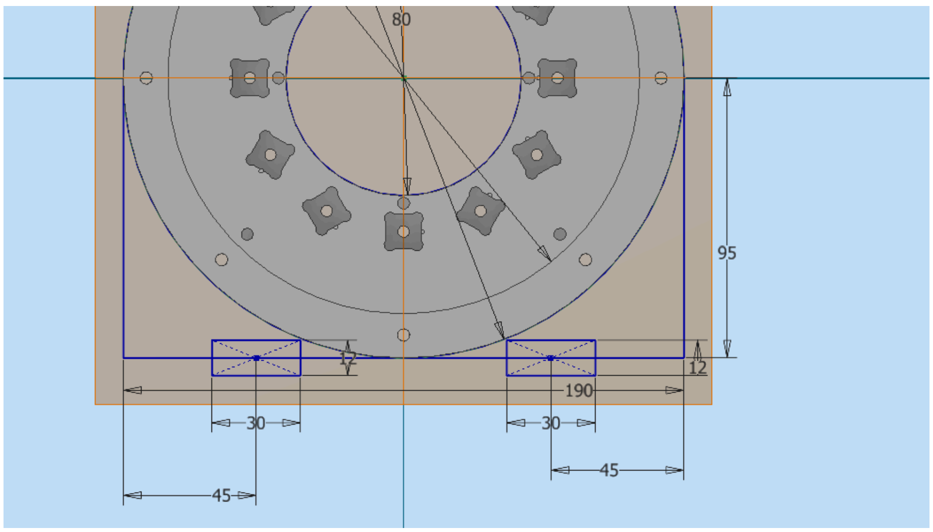
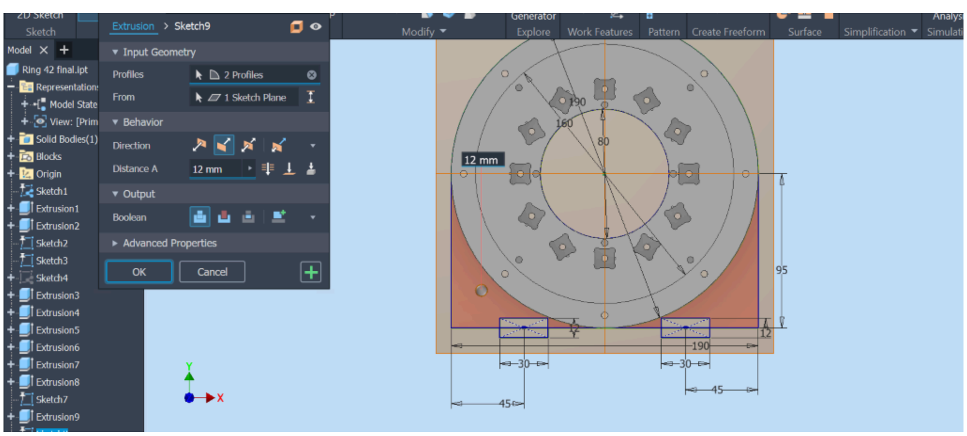
Direction
Distance A
12 mm
Output
Boolean
Advanced Properties
OK
Cancel


Ring 42 final.ipt
Representations
Model State: [Primary]
View: [Primary]
Solid Bodies(1)
Properties
Fillet
Selection Sets
Constant Radius Edge Sets
9 Edges
2 mm
Advanced Properties
OK
Cancel
Extrusion6
Extrusion7
Extrusion8
Sketch7
Extrusion9

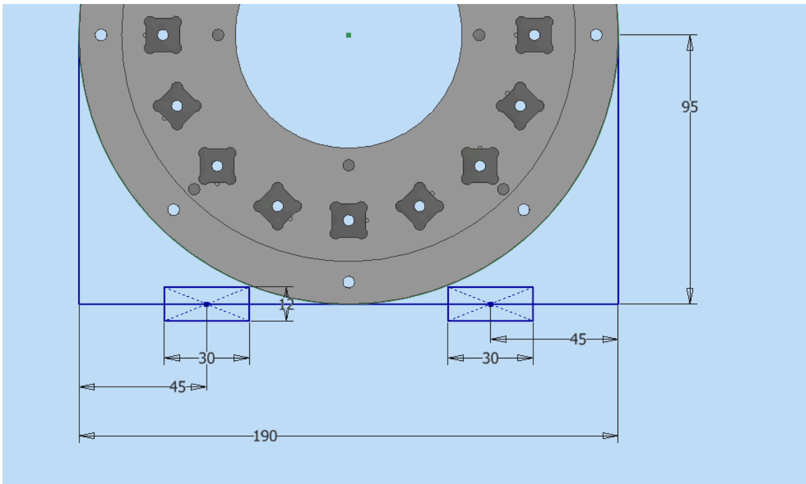




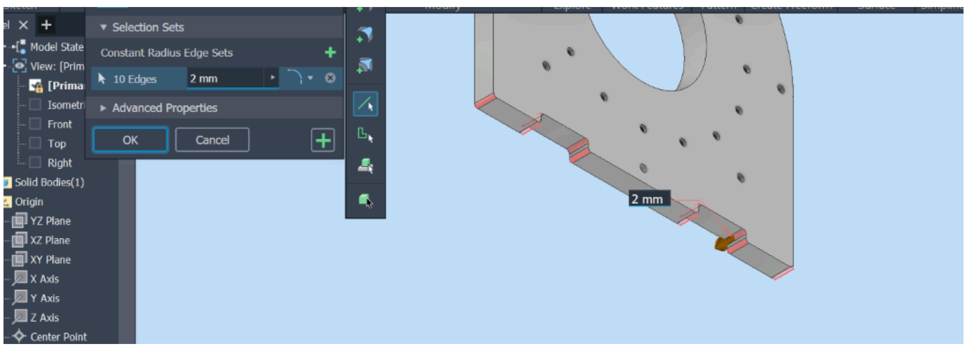
Extrusion



Ring 321



Fillet the edges



Steps to design Lids

Lid Design

Step 19 – Design Lids for Rings #288 and #251

The lids for Ring #288 and Ring #251 are designed using the corresponding ring dimensions.

1. Create a new sketch using the ring geometry as a reference.
2. Scale the lid features according to the requirements of the **120 mm inner bore diameter rat brain scanner**.
3. Extrude the support structure to a thickness of **12 mm**.
4. Apply fillets to smooth sharp edges and improve the final mechanical design.
5. Adjust design parameters as required and verify the final lid geometry.